

Lecture - 3

❖ Stages of Problem Solving

The six phases of problem-solving in C programming form a structured framework called the Program Development Cycle, which includes **Problem Definition, Problem Analysis, Algorithm Design, Coding, Testing/Debugging, and Documentation/Maintenance.**

1. Problem Definition

- This is the first step, where we clearly identify what problem needs to be solved.
 - ✓ Understand the main objective.
 - ✓ Define exactly what the program should accomplish.
 - ✓ Avoid misunderstanding the problem.
 - ✓ Set clear and specific goals.
- Example: Develop a program to calculate the area of a circle.

2. Problem Analysis

- In this stage, we determine what information is needed and what result should be produced.
 - ✓ Identify the inputs.
 - ✓ Identify the required outputs.
 - ✓ Determine data types such as int, float, or char.
 - ✓ Identify the mathematical formulas or rules required.
- Example:
Input: Radius (float)
Formula: $\text{Area} = \pi \times r^2$
Output: Area (float)

3. Design – Algorithm & Flowchart

- Here, the solution is planned before writing the actual code.
 - ✓ Develop a step-by-step algorithm.
 - ✓ Write pseudocode in simple English.
 - ✓ Draw a flowchart to visually represent the process.
 - ✓ Divide a complex problem into smaller modules.
 - ✓ Consider efficiency in terms of time and memory.
- Example:
Start → Read radius → Calculate area → Display area → Stop.

4. Implementation – Coding

- In this stage, the designed solution is converted into an actual C program.
 - ✓ Open a text/code editor.
 - ✓ Convert the algorithm or pseudocode into C syntax.
 - ✓ Use appropriate variables, functions, operators, and statements.
 - ✓ Add meaningful comments to make the code understandable.

- ✓ Save the program with a .c extension.
- Example: circle_area.c

5. Testing and Debugging

- After coding, the program must be compiled and tested.
 - ✓ Compile the C program.
 - ✓ Correct syntax errors.
 - ✓ Resolve header file and linking errors.
 - ✓ Test using different inputs.
 - ✓ Identify and fix logical and runtime errors.
- Example: Test the circle program with radius values such as 5, 10, 0, and negative values to check how the program behaves.

6. Documentation and Maintenance

- This is the stage where the program is explained, maintained, and improved after development.
 - ✓ Prepare user documentation or manuals.
 - ✓ Document important functions and code structures.
 - ✓ Fix bugs discovered later.
 - ✓ Modify or update the program when requirements change.
 - ✓ Improve performance and usability when necessary.